JOHNS HOPKINS
CENTER *for* TALENTED YOUTH

CTY LIVE STUDENT EVALUATION

Student: Aadya Agarwal Date: 7/22/2022

Course: Coasters and Corkscrews: Amusement Park Physics Instructor: Shawn Lemke

Congratulations, Aadya, on successfully completing Coasters and Corkscrews: Amusement Park Physics.

You mastered all course learning objectives and consistently exceeded expectations in class activities and assignments. In this course, you demonstrated a strong understanding of the core concepts and you effectively analyzed and applied these concepts to a wide range of issues and complex problems. In your contributions to class discussions and your written work, you successfully used your knowledge of facts and concepts to apply, analyze, and communicate understanding.

Throughout this course, you did a superb job of maintaining a fair and objective mindset when evaluating your own ideas and those of others. In addition, you frequently respected and fairly evaluated views, values, opinions, and approaches that were different from your own. You did an excellent job of taking initiative in this course and regularly developed original questions and sought answers on your own. You took ownership of your learning and were often able to be successful with very little support, a skill that will serve you well throughout your academic career and beyond.

Aadya, you displayed a strong understanding of the course concepts and applied them to building the roller coaster model. For example, in the Describing Motion Lab, you correctly calculated the average time, speed and the acceleration of the marble for various gradients. You did that by applying your knowledge of kinematics. You then wrote in that same lab “I think the relationship between the height of the ramp and the acceleration is that the higher the ramp the greater the acceleration.” This displayed your using the data and calculations to show how arrangements can be made to a structure to give the desired results. Also, for your final project of the roller coaster model, you then applied the very same concepts you learned in class to increase the size and gradient of the first hill. You stated “I would have a high drop and after the high drop because it is going so fast, a loopty loop would work, for my ride, I would have a lot of drops then add things such as loops and curves.” Aadya, these showed how you successfully synthesized the physics concepts to build the best roller coaster. Perhaps you will even think about these concepts the next time you are riding a roller coaster and enjoying yourself!

Aadya, it was a pleasure having you in class. Best of luck in the future!